EdPASS-H2O Grant: Year 3 Lessons Learned



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Dr. Robin Roberson, East Central University
Dr. Keith Strevett, University of Oklahoma (SME)
Grant #84034501

2024

Jessica Brumley, PhD

- Grant Project Manager
- Environmental Scientist
- Assistant Professor
- Director of Science Education
- Former high school science teacher



With special guest Nydia Lampkin

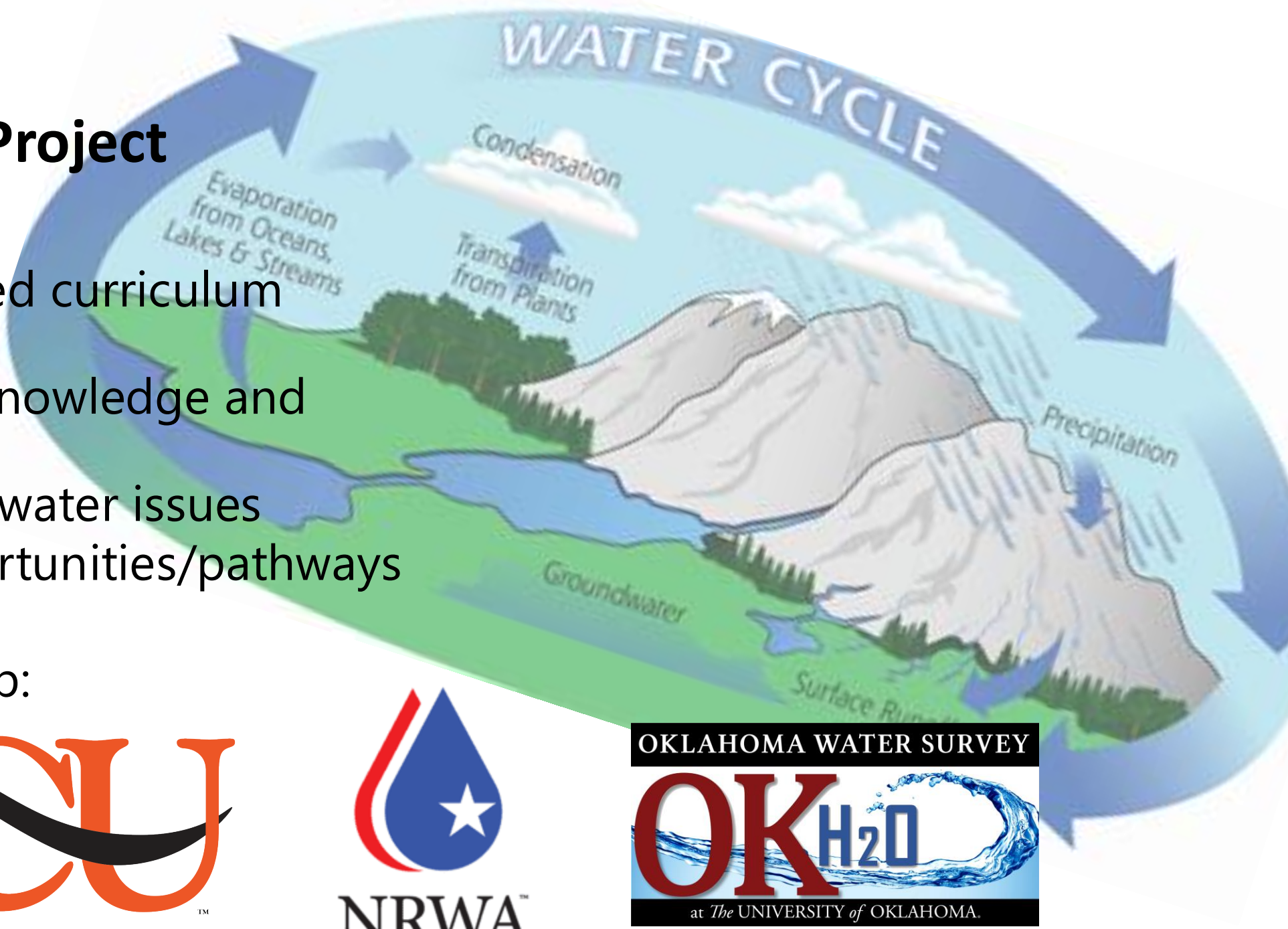
Robin Roberson, PhD

- Grant Curriculum Coordinator
- Educational Psychologist
- Associate Professor
- Former high school science teacher and high school counselor



EdPASS-H2O Project Description

- K12 STEM-based curriculum framework
- Foundational knowledge and awareness:
  - water/wastewater issues
  - career opportunities/pathways



Grant Partnership:

ECU

NRWA

OKLAHOMA WATER SURVEY
OK H2O
at *The* UNIVERSITY *of* OKLAHOMA

Goals:

1. Produce water consumers who understand both the natural water cycle and the human water cycle
2. Produce water consumers who are aware of and actively work to alleviate current and future water issues
3. Workforce development to produce water/wastewater infrastructure professionals prepared for this changing and challenging industry

EdPASS-H2O Roadmap

Introduce students to water and the water/wastewater industry

4th - 6th grade | 7th – 8/9th grade | 9/10th – 12th grade

Identity Development

- Local STEM fieldtrips
- Invited STEM speakers



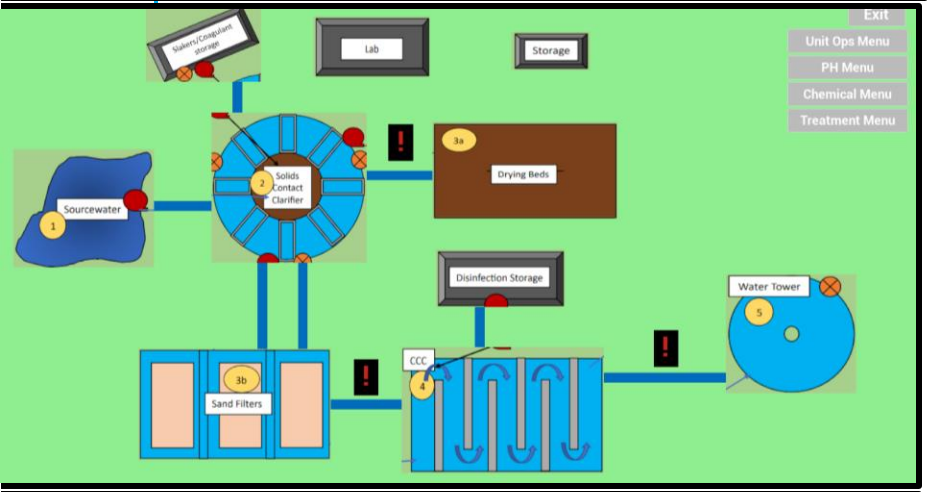
Career Identity Development

- Local STEM field trips
- Invited STEM speakers
- Career curriculum & field trips
- STEM traveling roadshow
- STEM day camps
- Problem/project-based learning modules

Career Development

- Local STEM field trips
- Invited STEM speakers
- Career curriculum & field trips
- STEM traveling roadshow
- STEM day camps
- STEM academies
- P/pBL modules



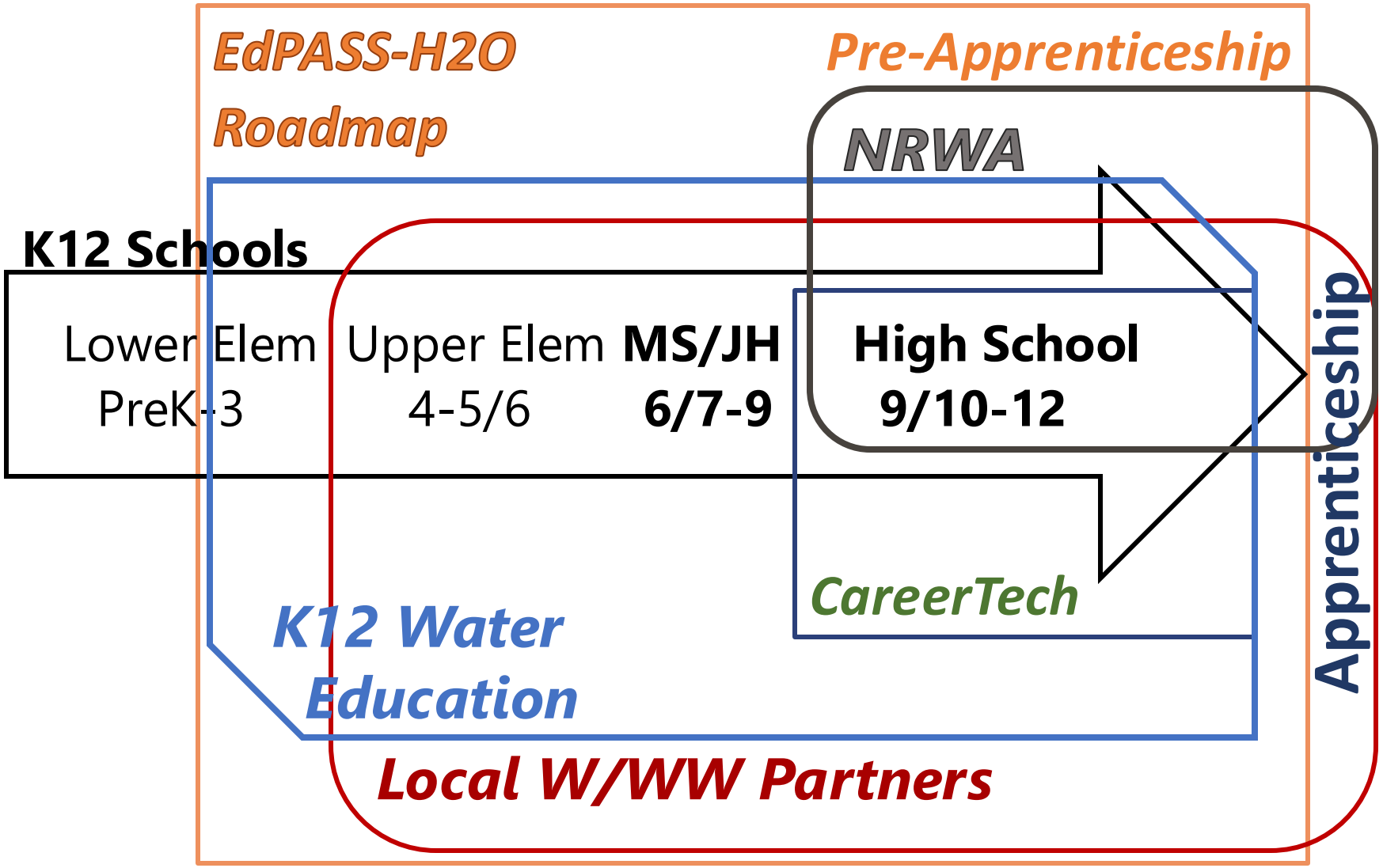




EdPASS-H2O Simulation

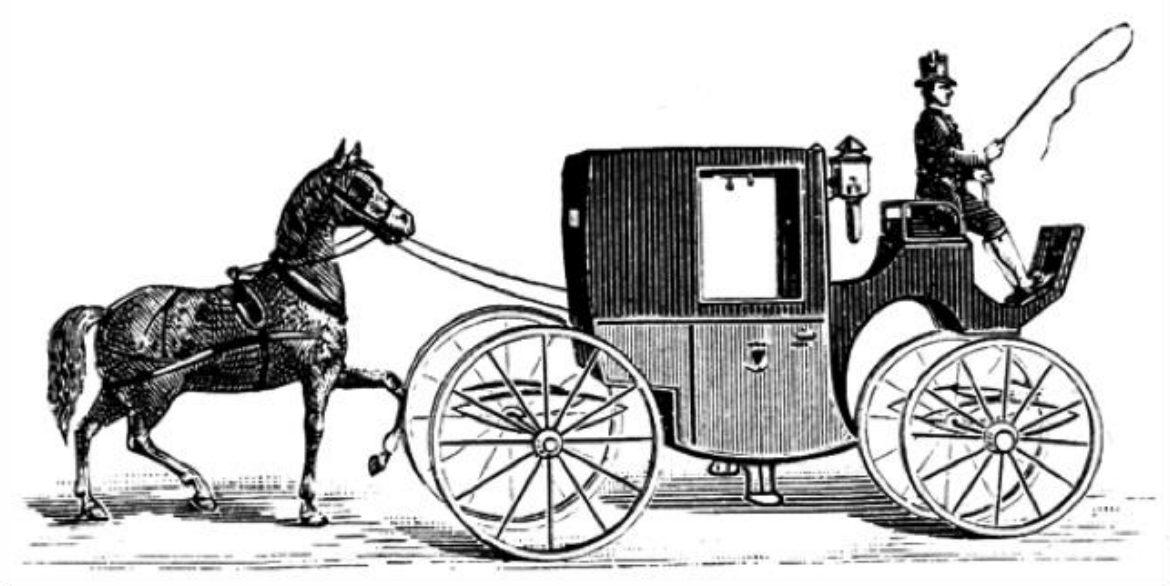
- Simulation ("game") to interest students in the water industry.
- Current beta environment is a water treatment plant.
  - Learn about the units within the simulation
  - Turn off valves
  - Replace pipes
  - Read turbidity
  - Watch educational videos
  - Answer quiz questions
  - Identify problems in the "SCADA"
  - Collect a sample
  - Measure pH

Current EdPASS-H2O Roadmap to Pre-Apprenticeship



Focus on industry integration with K12

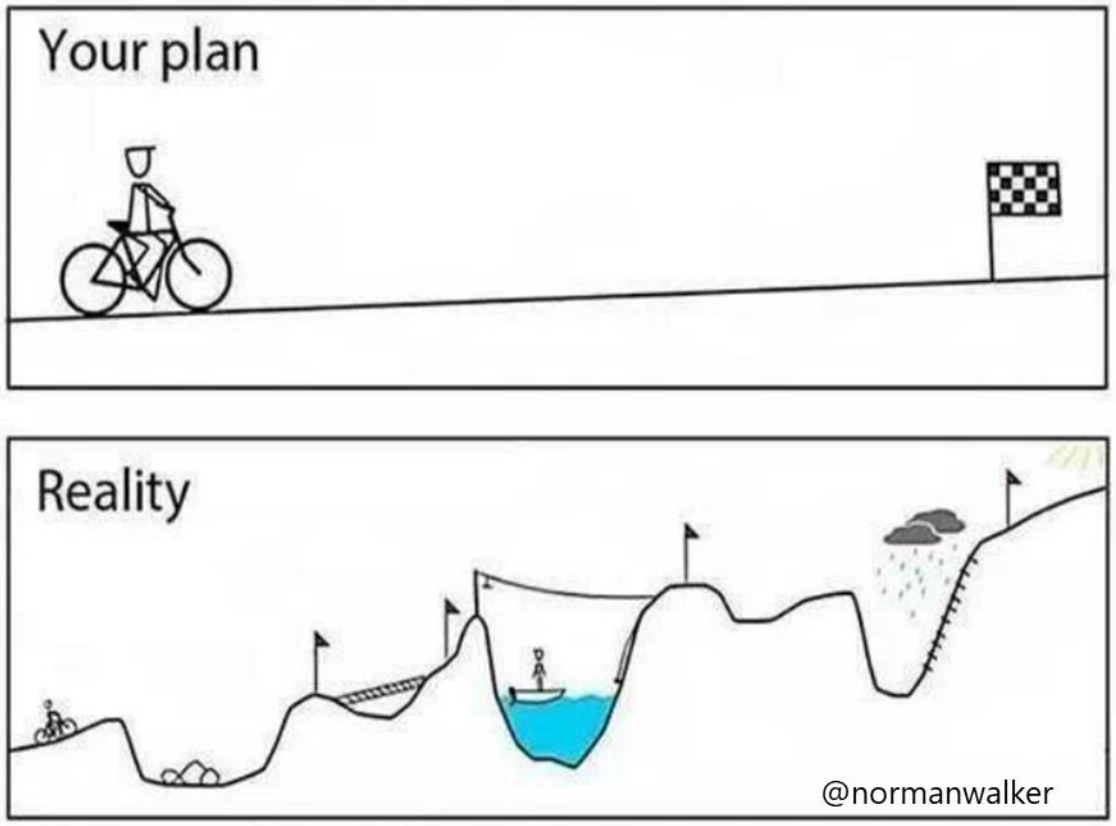
Cart before the horse…



Year 1: Outreach

- Tribal Nations
- Local Rural Water Districts
- Municipalities
- Career Tech
- State Level Entities
- Start on PBL Modules and Simulation App

Winding pathways…



Year 2: Connecting Dots

- K12 Outreach
  - Traveling Roadshow to STEM Academies: Water Science and Careers
  - Women in STEM (OKC, USAO)
- Career Tech
- State and local Municipalities/RWDs
  - OMAG
- K12 water education resources
  - PBL Modules and Curriculum
  - Google Classroom
  - Videos and Links
- Progress on the Simulation App

Zeroing in…



Year 3: Focusing

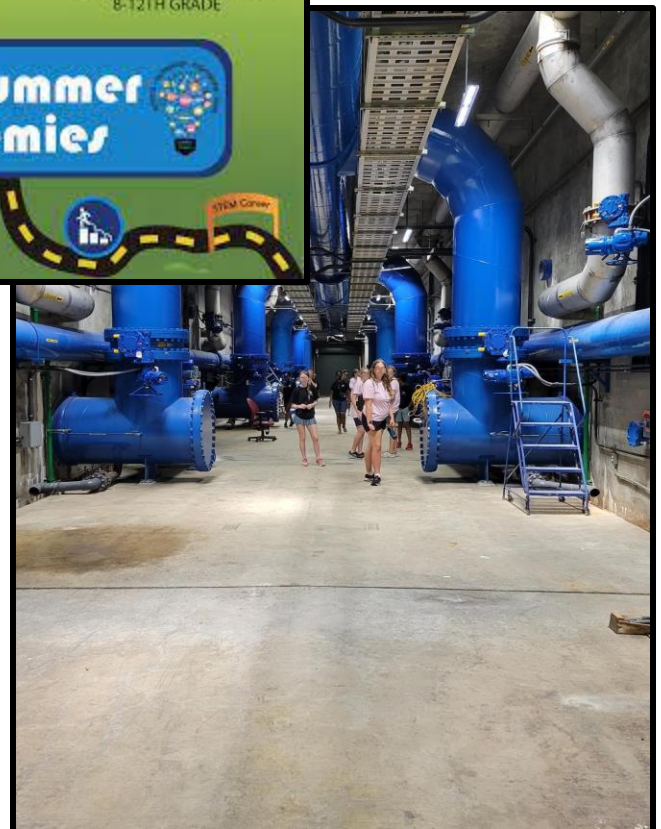
- Summer STEM Academies: *Human Water Cycle*
- K12 Outreach
  - *W is for Water*
  - *Tiger Academy*
  - *Women in STEM; STEAM Day; Women in Technology and Science*
- Progress on K12 water education resources
- Progress on the Beta Simulation
- National (WaterPro), regional (SWAWWA), state, and local water/wastewater partners
  - Share ideas and resources











Summer 2023: Traveling Roadshow

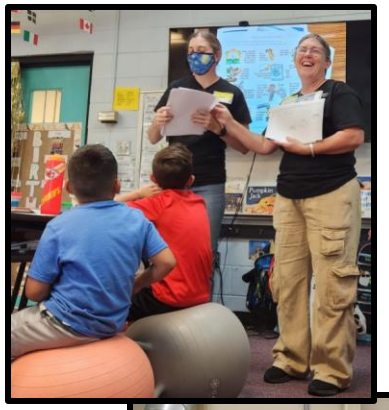
14 summer STEM academies across Oklahoma

- 2-4 hours each visit
- Multiple activities and labs:
  - Treatment plant tours
  - Microscopes: Living water
  - Coagulation and flocculation
  - Filtration
  - Water quality testing
  - Engineering challenge
  - Secchi disc build
  - Careers exploration
  - App simulation

K12 Outreach Fall 23/Spring 24

W is for Water – Ada Public Schools

- Taught in-class water curriculum at school sites:
  - Grade 3 (water cycle)
  - Grade 8 (water quality)
  - Grade 11/12 APES (careers)



- Took APES class on local water tour
  - ECU Center for Undergraduate Research Lab
  - Ada Wastewater Treatment Plant
  - Birds Mill Spring (Ada water source)
  - EPA Lab (RS Kerr Environmental Research Center)



K12 Outreach Fall 2023

Women in STEM at Science Museum of OK (EPSCoR)

- 6-12 grade (water testing, scopes, EdPASS app)

STEAM Day at ECU (voted favorite session)

- 6-12 grade (water testing, scopes)

Women in Technology and Science USAO

- 5-6 grade (water testing, scopes, careers)





K12 Outreach Spring 2024

Tiger Academy

- OSRHE STEM grant
  - After-school program
  - Grades 4-7 (6th focus)
  - Math *remediation*
  - Science/engineering/ technology *enrichment*
    - Water conservation



Summer 2024 OSRHE STEM Academies: The Human Water Cycle

- STEM Academy focused on drinking water and wastewater
- Same academy – three sites



Rose State College
Midwest City, OK



East Central University
Ada, OK

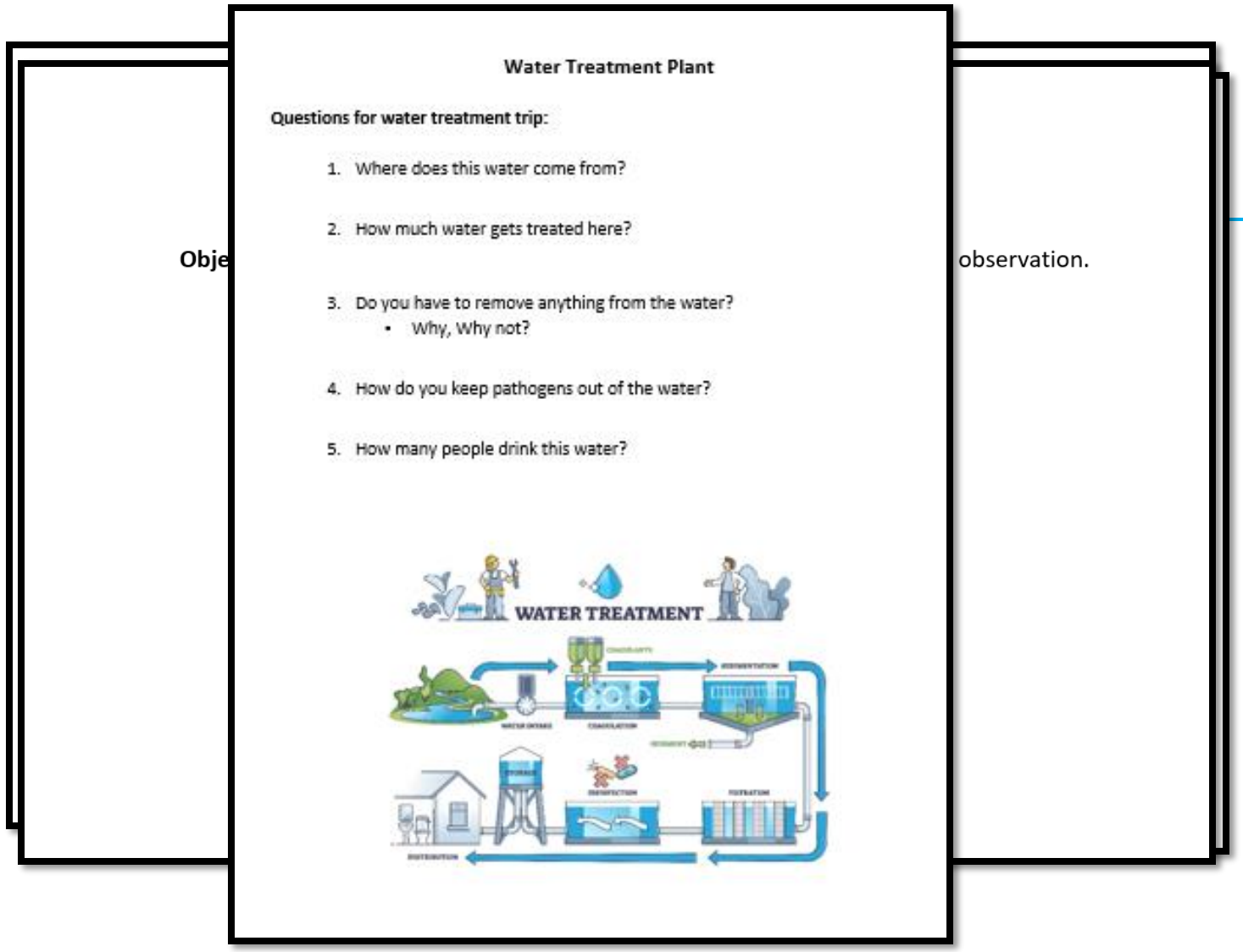


University of Oklahoma
Norman, OK

Day 1: Water Quality & Resource Management

Schedule

- Lab Notebook



Day 1: Water Quality & Resource Management

Schedule

- Collect water samples
  - Water testing





Day 1: Water Quality & Resource Management

Schedule

- Collect GPS data
  - Create GIS map
- Drones

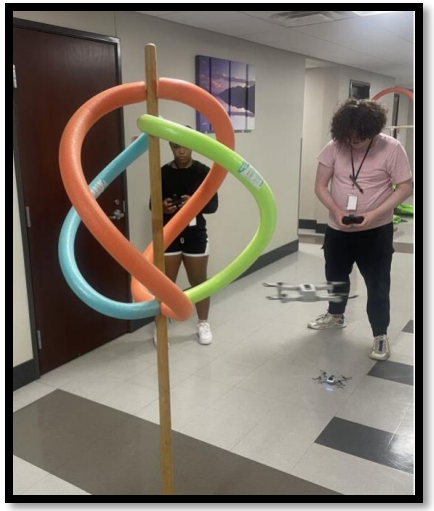
Flight Rubric

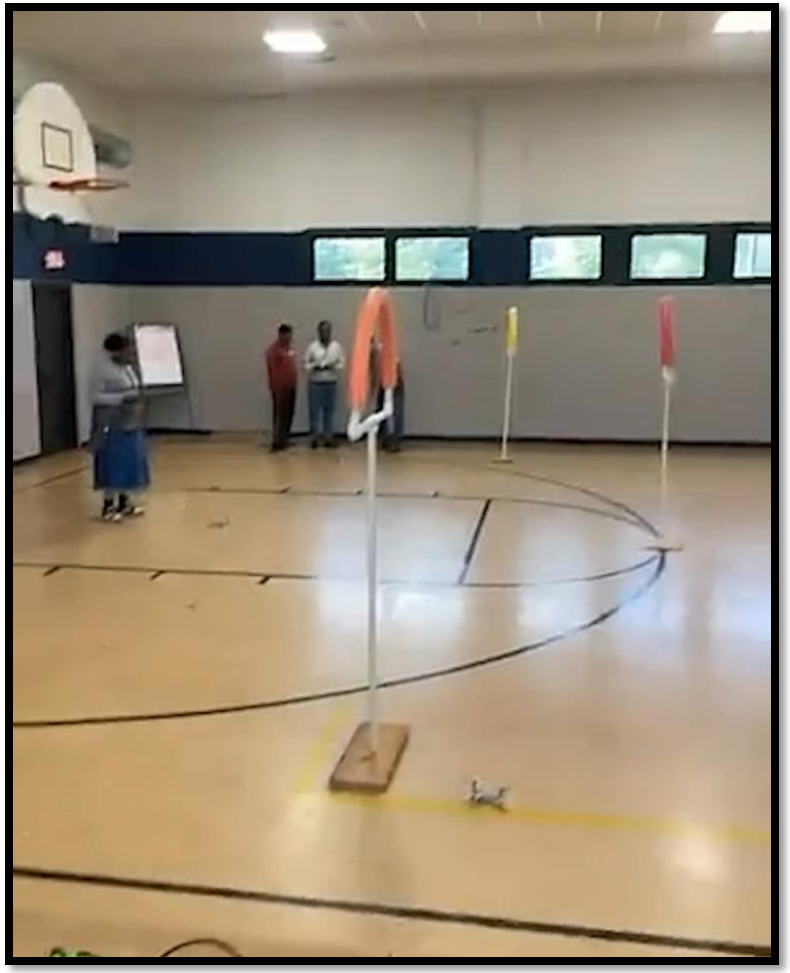
Teammate 1

Knowledge/Skills	Points
Team communication	/10
Launch	/1
Flight – land on target (points awarded based on rings on target)	/5
Flight – fly through hoop	/3
Flight – successfully fly three obstacles in a row	/10
Controlled landing	/4
Stuck in net (each time)	/-1
Hit obstacle (each time)	/-1

Teammate 2

Knowledge/Skills	Points
Team communication	/10
Launch	/1
Flight – land on target (points awarded based on rings on target)	/5
Flight – fly through hoop	/3
Flight – successfully fly three obstacles in a row	/10
Controlled landing	/4
Stuck in net (each time)	/-1
Hit obstacle (each time)	/-1

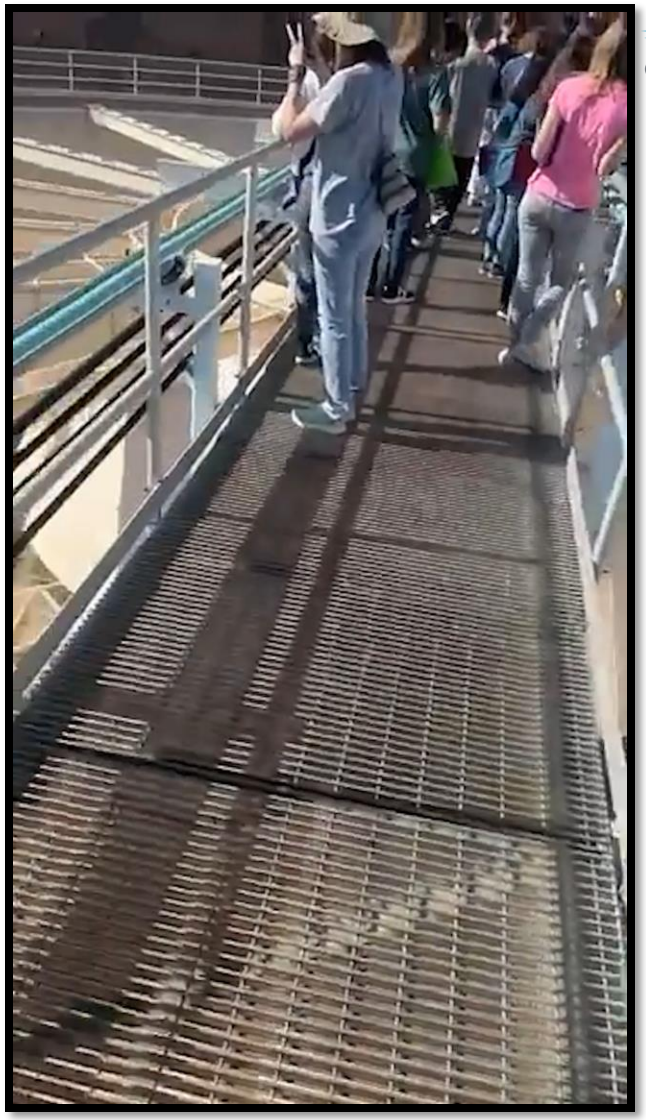




Day 2: Water Treatment

Schedule

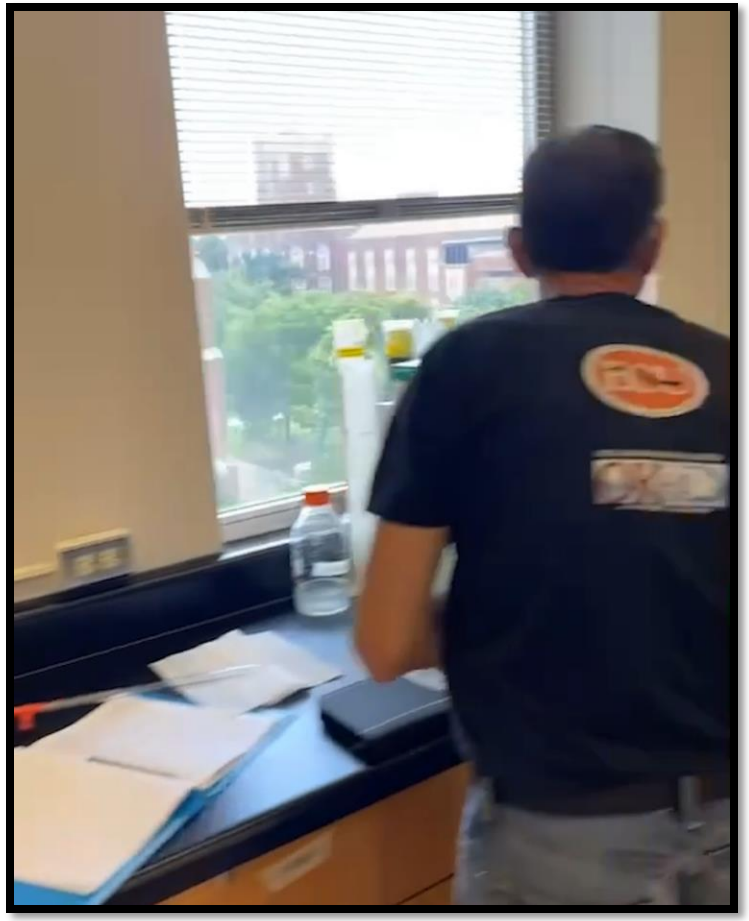
- Tour local water treatment plant
- WatPro treatment plant simulation



Day 2: Water Treatment

Schedule

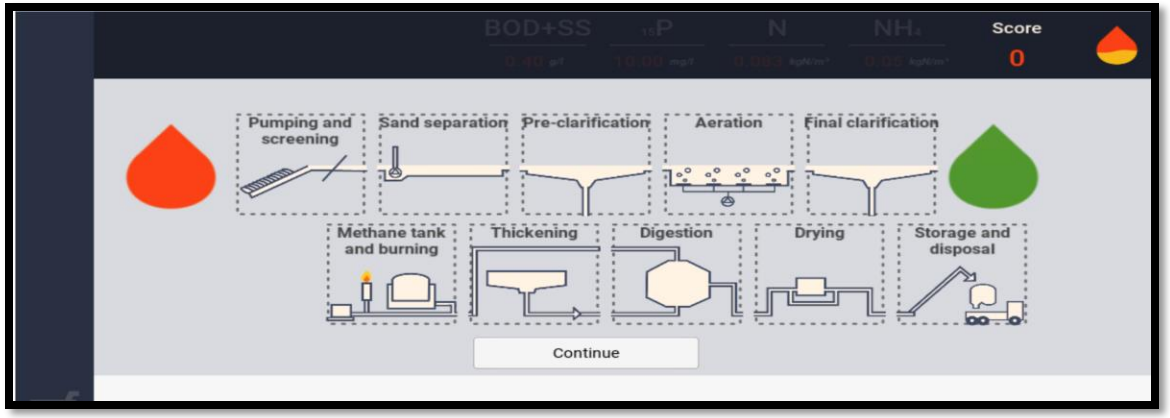
- Coagulation experiment
- Flocculation & sedimentation experiment

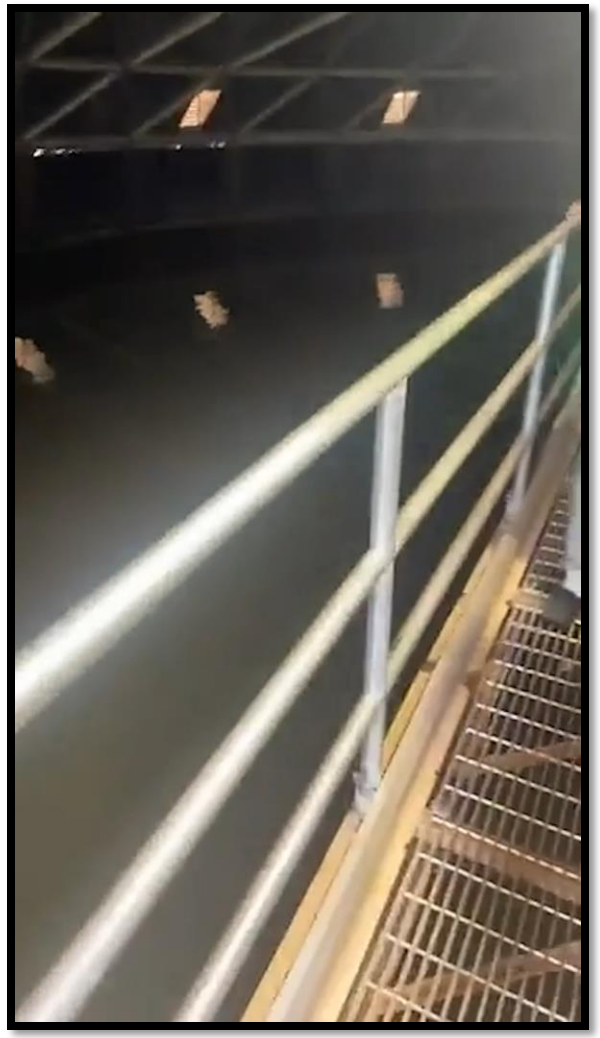


Day 3: Wastewater Treatment

Schedule

- Tour local wastewater treatment plant
- Baltic Smart Water Simulation

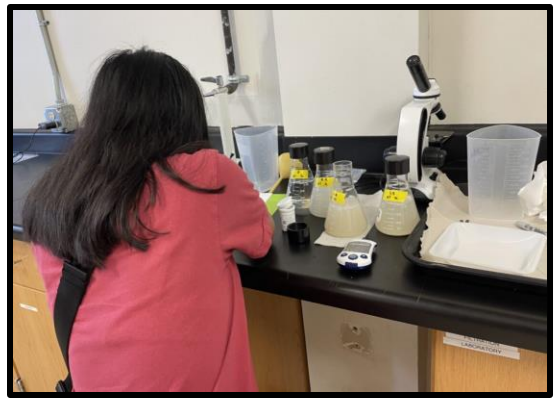




Day 3: Wastewater Treatment

Schedule

- Glucose reduction experiment
- Gram staining & colony counts
- Secondary biological treatment & microscopy

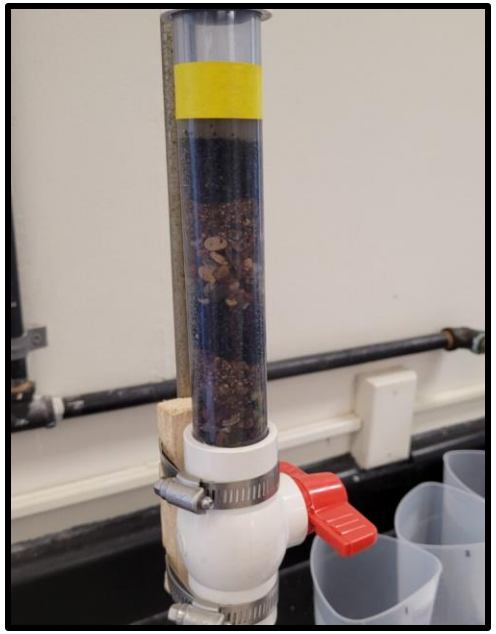




Day 4: Engineering Challenge

Schedule

- Introduction to Sand Filtration
- Student experimentation with media
- Student build & test filter
- Student presentations





Traveling Roadshow vs STEM Academy

Roadshow

- Take resources and go to students
  - Preparation
- Specific STEM/Water Activities
  - Not very in-depth
- Less time and cost required
  - Do activity & leave
  - Possible STEM grant funding
- Student behavior is host's responsibility
- Students enjoy the activities
  - Smaller effect on student attitudes

Academy

- Students come to you & resources
  - Preparation
- Specific STEM/Water Activities
  - More in-depth
- More time and cost required
  - Residential, commuter, afterschool
  - Likely STEM grant funding
- Student behavior is your responsibility
- Students enjoy the activities
  - Larger effect on student attitudes
    - Relationship building

What can you do?

Advertise plant tours to the surrounding communities

- Work with educators to make tours age-appropriate and surprise-free

Traveling Roadshow

- Contact local schools or teachers you know
- Provide/lead in-class water science activities

Afterschool Program or Academy/Camp

- *Reach out to* and partner with a local tech center, community college, university
  - You provide technical expertise
  - They provide teaching expertise

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